IMPACT TANK ’26 · MARKET VALIDATION + GO-TO-MARKET

The Journey After Harvest

Connecting traceability, operational coordination and smarter produce-distribution decisions

Tomato · Nagpur orange · Onion | Nagpur Division, Maharashtra



[Team name] · [Members] · [Institution]

Simulation output, not field-validated · personas are hypothesised

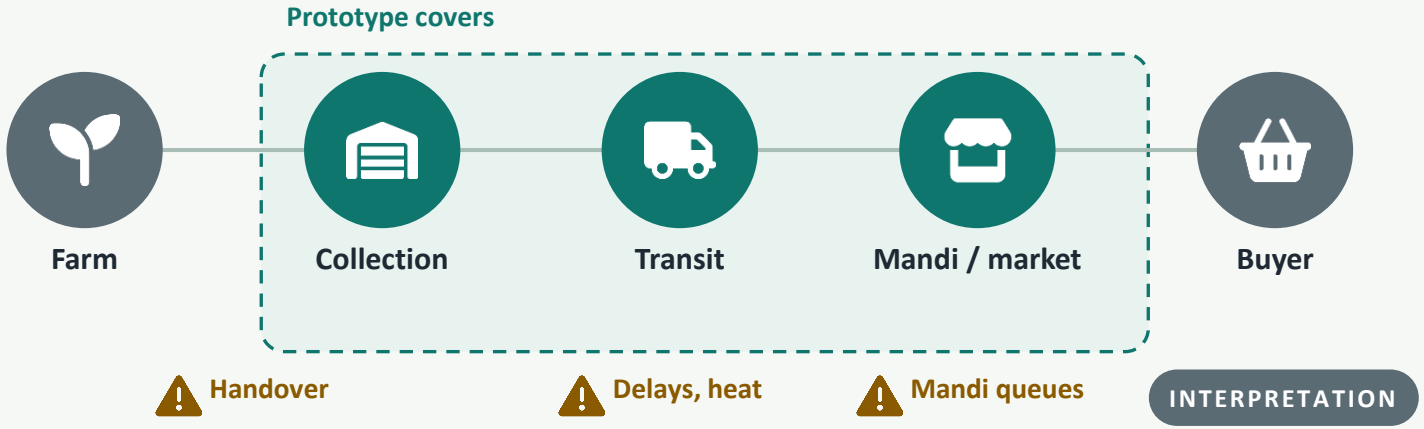
Large losses, unknown transit share

19.33 Mt

of fruits (7.36 Mt) and vegetables (11.97 Mt) lost after harvest

DOCUMENTED

NABCONS study (2022) for MoFPI, 54 crops, 2020–22 · PIB, 1 Aug 2025



Our crops: % lost (NABCONS)

Crop	Farm level	Market level
Tomato	8.37%	3.25%
Citrus (incl. orange)	5.53%	2.18%
Onion	5.31%	1.96%

Transit is not reported separately

Transit-stage share: unknown

HYPOTHESIS TO VALIDATE

NABCONS reports farm-level and market-level loss; what happens on the road is not measured separately. Closing that gap is the point of the operations layer.

Quality falls with time and temperature

INTERPRETATION

So when and where a load goes matters. A published production-routing model treats quality as a function of time and temperature.

Manouchehri, Shahandeh Nookabadi & Kadivar, Heliyon 2020 (models chicken meat; we use the principle only)

One supply chain, five different needs

HYPOTHESIS TO VALIDATE



Farmer

PAIN

Little record of the weight handed over

IN THE APP

Gets a QR label; both sides confirm the weight

HYPOTHESISED



Collection-point supervisor

PAIN

Handovers, wastage and crowding tracked by hand

IN THE APP

Scans lots, logs wastage, sees shelf-life lights

HYPOTHESISED



Grocery store owner

PAIN

Supply and condition don't match demand

WOULD DO (FUTURE)

Future: a view of incoming supply and shelf life

ROADMAP



Truck driver

PAIN

Blamed for spoilage that happened elsewhere

IN THE APP

Two-party handover record at pickup and drop-off

HYPOTHESISED



NGO owner

PAIN

Short notice on surplus; pickup is hard to plan

IN THE APP

Sees nearby offers; claims within 2 hours

HYPOTHESISED

Persona 1 · Farmer

HYPOTHESISED PERSONA *No direct farmer interviews in this window*

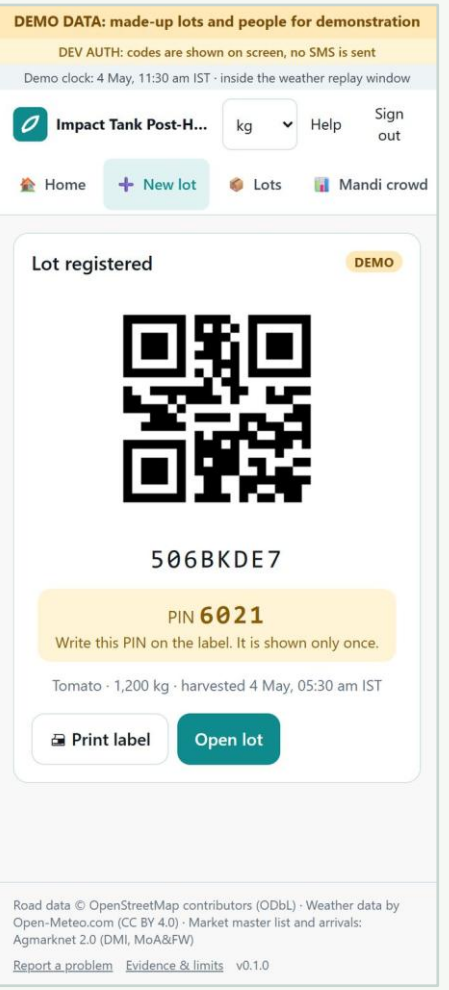
“Same mandi, same buyer, plus a weight record.”

CONTEXT	Brings tomato, orange or onion to a mandi or collection point
GOALS	Know how much was handed over, and to whom
PAINS	Little accessible record of the handover and the recorded weight
WE OFFER	QR-labelled lots; two-party handovers in a tamper-evident log

ASSUMPTION TO TEST HYPOTHESIS TO VALIDATE

Farmers value a digital handover and weight record enough to use it regularly





Prototype: QR label + PIN (demo data)

Persona 2 · Collection-point supervisor

HYPOTHESISED PERSONA *Direct field validation pending*

“A red/amber/green shelf-life light and simpler lot tracking.”

CONTEXT Receives incoming lots and confirms handovers at a collection point

GOALS Organised lot records, visible activity, recorded wastage

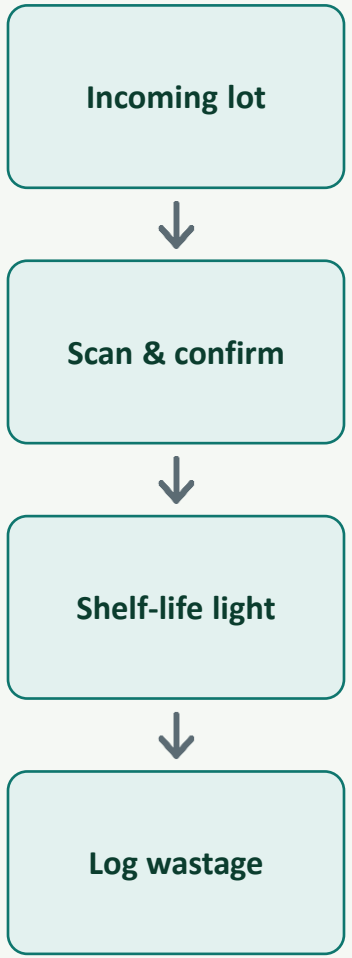
PAINS Handovers, wastage and congestion recorded inconsistently

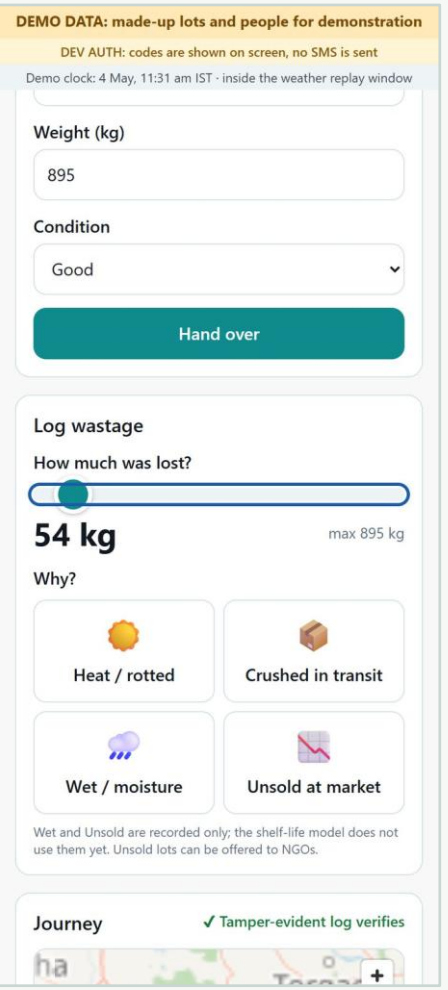
WE OFFER QR lots · two-party handovers · one-tap wastage logger · shelf-life lights · crowd meter · Route Advisor

ASSUMPTION TO TEST

HYPOTHESIS TO VALIDATE

One consolidated interface makes daily collection-point management more useful





Prototype: wastage logger (demo data)

Persona 3 · Grocery store owner

ROADMAP SEGMENT *Retailer functionality is a roadmap item*

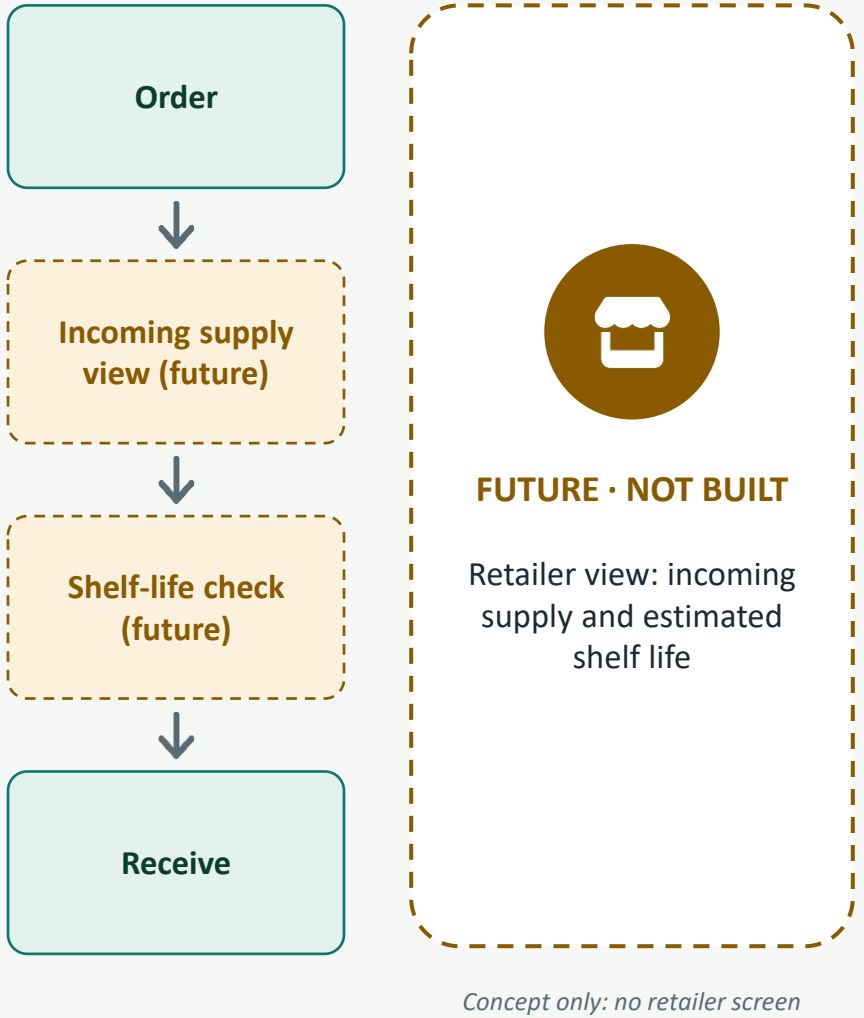
The prototype has NO retailer screens yet. This is a buyer-side segment for the roadmap.

CONTEXT	Buys produce from wholesalers or distribution points
GOALS	The right quantities, with enough usable life left
PAINS	Incoming supply and condition don't match expected demand
WE'D OFFER (FUTURE)	A retailer view of incoming supply and estimated shelf life

ASSUMPTION TO TEST

HYPOTHESIS TO VALIDATE

Retailers value better visibility of produce condition and incoming supply



Concept only: no retailer screen

Persona 4 · Truck driver

HYPOTHESISED PERSONA *Simulation-based; not field-validated*

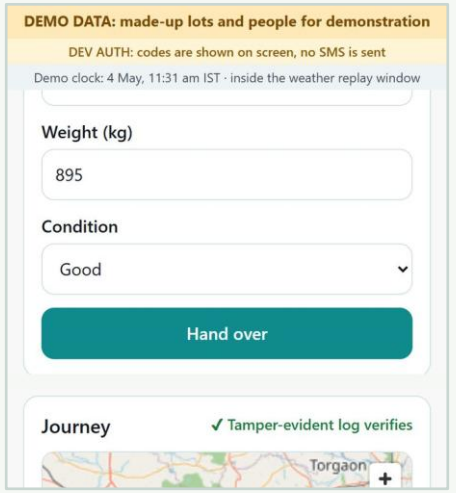
"The Route Advisor recommends. It does not automatically reroute trucks."

CONTEXT	Moves produce between collection points, mandis and markets
GOALS	Deliver on time with quality intact, without unfair blame
PAINS	Delays and disruption; blame for spoilage that happened elsewhere
WE OFFER	Two-party handover record at pickup and drop-off; a dispatcher-confirmed route

ASSUMPTION TO TEST HYPOTHESIS TO VALIDATE

Drivers and dispatchers find the handover record and route recommendations clear and useful





Prototype: handover + log check (demo)

Persona 5 · NGO owner

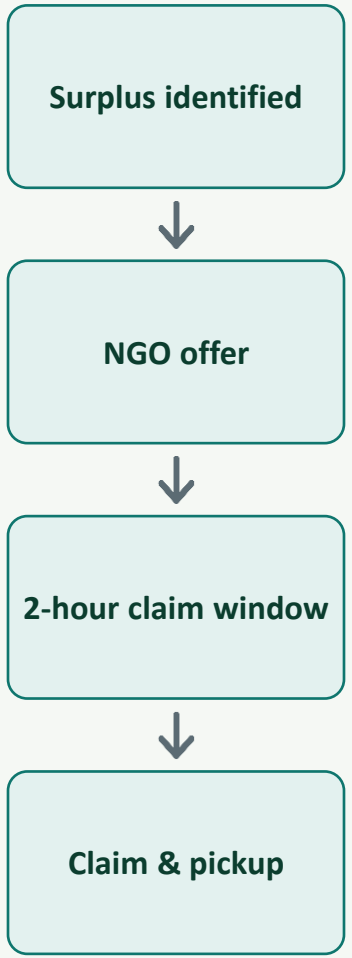
HYPOTHESISED PERSONA *Adoption and rescue not field-validated*

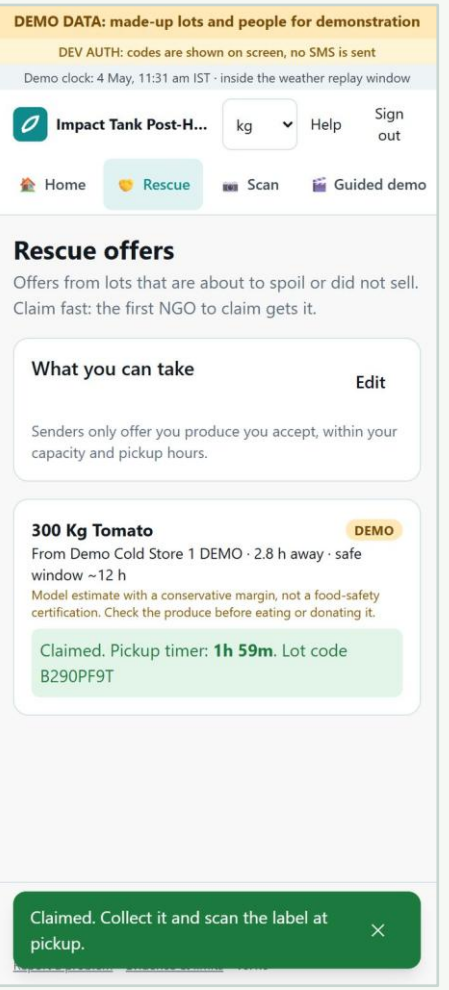
“Offers for nearby surplus, with a 2-hour claim timer.”

CONTEXT	Can receive suitable surplus produce for redistribution
GOALS	Spot usable offers early and decide in time
PAINS	Short notice, uncertain availability, pickup coordination
WE OFFER	NGO rescue offers with a 2-hour claim timer

ASSUMPTION TO TEST HYPOTHESIS TO VALIDATE

NGOs would monitor, claim and coordinate eligible offers within the window





Prototype: NGO claim timer (demo data)

What we tested, and what we haven't

VALIDATION STATUS

In the short window available we could not get time with working farmers or dispatchers. Farmers were unavailable at short notice, and dispatchers and commission agents were difficult to reach quickly. So this round's evidence is mainly published research and expert/second-hand input, plus [PLACEHOLDER: number and type of any calls or surveys we did complete]. Our next step is direct interviews, listed on the roadmap slide.

Assumption	How tested	Evidence	Status
QR lots and two-party handovers are a usable workflow	Automated end-to-end tests	Software works; no user trials yet	**Untested**
One-tap wastage logging fits daily work	Automated tests	Software works; no user trials yet	**Untested**
Shelf-life lights and crowd meter are useful signals	Not yet tested with users	[PLACEHOLDER]	**Untested**
NGOs can act within a 2-hour claim window	Not yet tested with NGOs	[PLACEHOLDER]	**Untested**
Quality-aware routing improves outcomes overall	Pre-registered simulation, 3 seasons	No overall winner (next slide)	**Unsupported**
Dispatchers and traders would adopt it	Not yet tested	[PLACEHOLDER]	**Untested**
Route recommendations are commercially useful	Not yet tested	[PLACEHOLDER]	**Untested**
Stakeholders would pay	Not yet tested	[PLACEHOLDER]	**Untested**

INTERPRETATION Published modelling (Manouchehri et al., 2020) supports treating quality as time- and temperature-dependent. It is not evidence that our system works.

What the simulation taught us, and what changed

Pre-registered simulation result

DOCUMENTED

Overall

No overall winner

Summer wastage

Lower

Monsoon wastage

Lower

Unmet demand

Higher in every season

Same lots, weather and roads; only the routing policy differs. 20 seeds per season. Simulation output, not field-validated.

Changes already made

Route Advisor: “The app recommends; you decide.”

Each result carries a caution: in simulation this policy cut waste in hot seasons but raised unmet demand. Both metrics are side by side in the Simulation Lab.

DOCUMENTED

First users join a staffed demo and pilot waitlist

Because sign-in is still a development stub.

DOCUMENTED

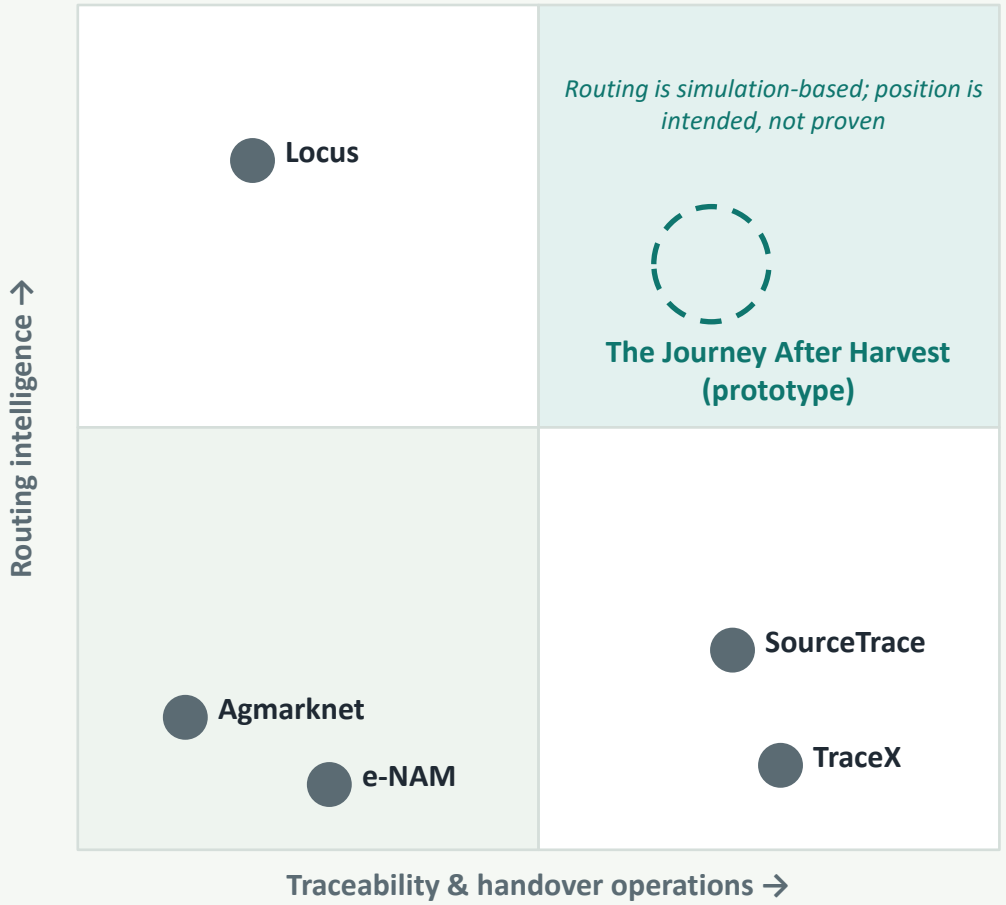
To investigate next

HYPOTHESIS TO VALIDATE

- Show the trade-off inside the advisor itself
- Clearer shelf-life view · simpler records for early users
- Human approval before any routing decision

Field-interview insights: [PLACEHOLDER]

Where we sit among existing tools



Product	Documented on its official site	Traceability	Routing
Agmarknet	Govt portal: mandi prices and arrivals	Not described	Not described
e-NAM	Govt online agri-trading platform; has an e-NAM-Logistics section	Not established	Not established
SourceTrace	Farm-to-retail traceability and supply-chain SaaS	Yes	Not described
TraceX	Blockchain traceability, EUDR compliance	Yes	Not described
Locus	Route planning, dispatch and shipment tracking	Not described	Yes
Ninjacart	Fresh-produce B2B marketplace, farmers to buyers	Not described	Not described

Placement from each product's public website, reviewed 28 Oct 2026. Ninjacart not placed: not established from documentation.

DOCUMENTED

"An operational traceability and decision-support prototype designed around the journey of perishable produce after harvest."

We don't claim to be first, only or best.

Who might pay: hypotheses, not prices

FREE · hypothesis

Farmers

Free access

Price: none

WE'LL ASK

"How do you keep a record of what you hand over today, and what would make you keep using a record like this?"

NGOs

Free access

Price: none

WE'LL ASK

"How do you learn about surplus produce today, and what would you need to claim it within two hours?"

PAID · hypothesis

Collection centres

Tier: lot records, handover history, monitoring

Price: [PLACEHOLDER]

WE'LL ASK

"What do lot records and wastage tracking cost you today, in time or money, and what would you pay to change that?"

Dispatchers

Tier: route comparison and trade-off view

Price: [PLACEHOLDER]

WE'LL ASK

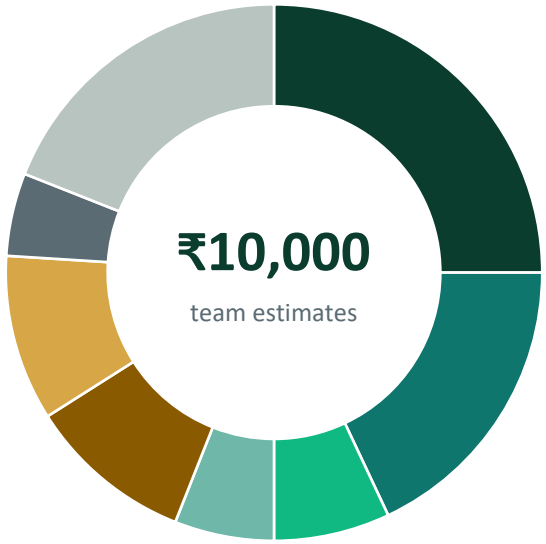
"How do you choose a market for a load today, and what would you pay for help comparing options?"

Later: larger operators · multi-location deployments and integrations · Price: [PLACEHOLDER] · We'll ask: "Would one system across several locations change how you run operations?"

HYPOTHESIS TO VALIDATE

Willingness to pay ≠ demand

How ₹10,000 reaches our first users



Travel	₹2,500
Posters + banner (Marathi/Hindi)	₹1,800
Handouts	₹700
300 QR stickers	₹600
Demo desk	₹1,000
Refreshments	₹1,000
Mobile data	₹500
Reserve	₹1,900

Four field days

- Visual posters per persona, in Marathi and Hindi
- Staffed demo desk near commission-agent shops (APMC permission first)
- Partner-led demos via FPOs, KVKs and NGOs

If APMC permission is delayed: digital-first (WhatsApp demo videos, QR-linked info, waitlist) or a partner-led split via FPOs/KVKs.

Demo targets

TARGETS, NOT RESULTS

15	8	3	3
farmers / FPO members	dispatchers	supervisors	NGOs

Reach → Demo → **Activation** → Pilot

Activation = a completed demo handover or a pilot-lot commitment. Not sign-ups or QR scans.

First users join through a staffed demo and pilot waitlist. Prototype sign-in is a development stub.

Simulation output, not field-validated · personas are hypothesised

From prototype to pilot

NOW

- Direct interviews: farmers, dispatchers, commission agents, supervisors, NGOs
- Staffed pilot at 2–3 collection centres in Nagpur Division
- Real operational feedback

NEXT

- Retailer view (grocery store owner)
- SMS / WhatsApp notifications
- Postgres persistence and real authentication
- Refine from pilot evidence

LATER

FUTURE · NOT BUILT

- Sensors in trucks and transit points
- Learned machine-learning models
- Wider deployment

"Our next milestone is not more technology. It is proving that the right information, at the right handover, can help make better decisions after harvest."

Simulation output, not field-validated · personas are hypothesised